

MANUEL BUENROSTRO-MACEDO

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Engineering Portfolio: <https://buenrosm23.github.io/>

EDUCATION

Embry-Riddle Aeronautical University — Daytona Beach, FL
Bachelor of Science in Aerospace Engineering, Rocket Propulsion
Expected Graduation: May 2028 | GPA: 3.0 / 4.0

ENGINEERING EXPERIENCE

Engineering Physics Propulsion Lab (EPPL) - Embry-Riddle Aeronautical University
Undergraduate Research Assistant | 2024 - Present

- Integrated and tested lab hardware, expanding research project hardware capabilities.
- Presented technical solutions to sponsors, visitors, and faculty at lab showcase events.
- Applied engineering principles to create, revise, and improve mechanical parts in Autodesk Inventor.
- Gained firsthand experience in a professional research lab setting.

TECHNICAL PROJECTS

OpenArms – Open-source Robotic Arm - EPPL, ERAU | January 2025 - Present

- Team member in the design of an open-source robotic arm attachment for OpenMutt.
- Developed interchangeable end-effectors with CAD and additive manufacturing, allowing for object manipulation and environmental interaction on the OpenMutt-OpenArms platform.

OpenMutt – Open-source Quadraped - ERAU | September 2024 – Present

- Developed pan-tilt multi-sensor mount using CAD and additive manufacturing expanding robotic quadraped sensing and environmental awareness.
- Designed and fabricated a quick disconnect mounting system for interchangeable end-effectors extending quadraped functionality using additive manufacturing and Autodesk Inventor.

VEX Robotics Competition - Team 7121E | March 2020 – May 2024

- Led 7-member team to 21 regional, 3 state, and 1 world championship appearances; won 7 design-merit awards.
- Team leader, head designer; built award-winning robots with pneumatics and CAD; maintained detailed engineering documentation.

TECHNICAL & PROFESSIONAL SKILLS

Software & Tools: Autodesk Inventor, CATIA V5, MATLAB (basic), Python (beginner), 3D printing, laser cutting, CNC (beginner), Windows Ecosystem, Visual Studio Code (Beginner), Prusa Slicer, Bambu Studio.

Soft Skills:

- Leadership, teamwork, iterative design, technical presentations, technical report writing, documentation
- Experience with shop tools, inventory, and safety procedures.
- Languages: Native in English and Spanish